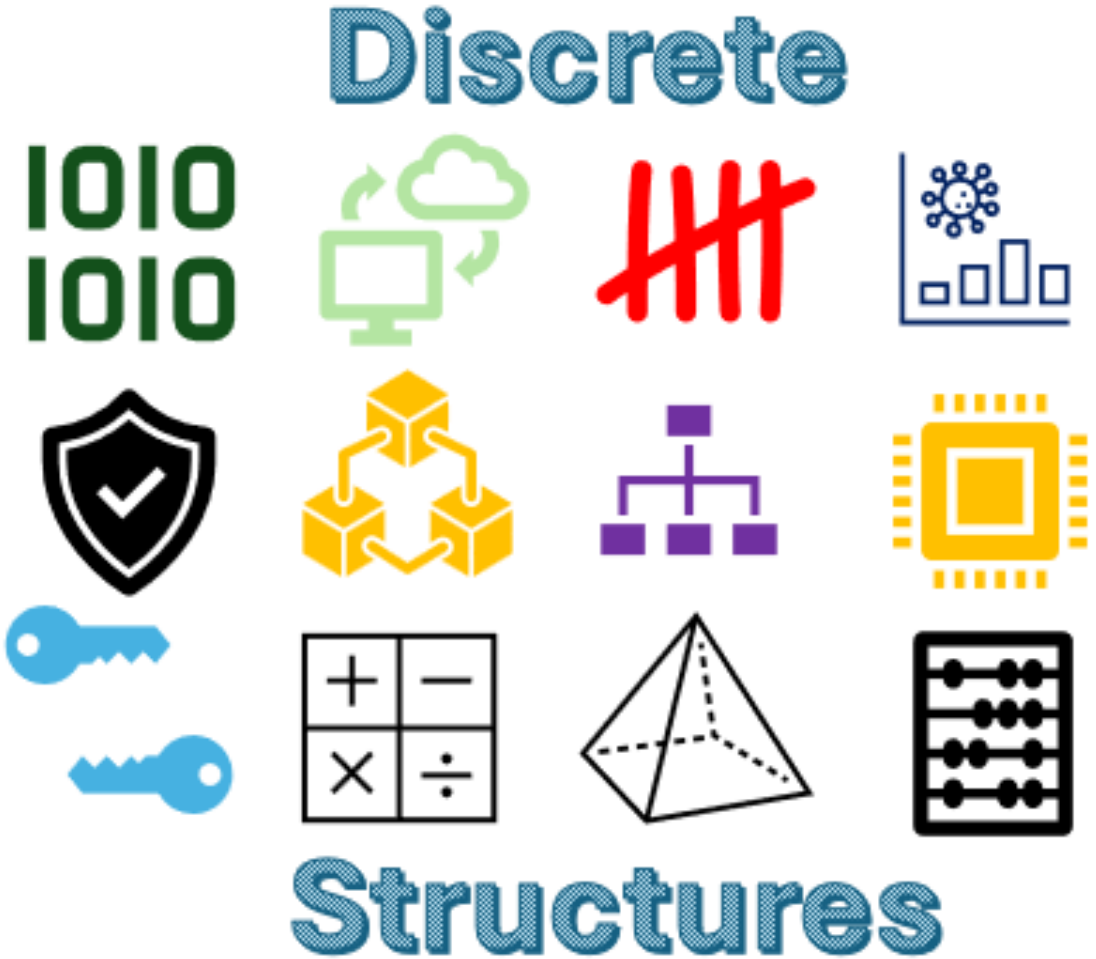
COMP 2310: Discrete Structures
Fall 2026

Chapter 1: Fundamentals



Greatest Common Divisor

Euclid(m, n)
//Computes gcd(m, n) by Euclid's algorithm
//Input: Two nonnegative, not-both-zero integers m and n
//Output: Greatest common divisor of m and n
while n ≠ 0 **do**
    r ← m mod n
    m ← n
    n ← r
return m

Can you solve the problems below step-by-step?

gcd (20, 5) = ?

gcd (4, 16) = ?

Least Common Multiple (LCM)

- The smallest positive number that is a multiple of two or more given numbers.
- The LCM of two or more numbers is always greater than or equal to the largest of the numbers.

Obtain LCM from GCDc

$$\text{LCM(M, N)} = \frac{|M \times N|}{GCD\ (M,N)}$$

Matrices: few requirements

- **Matrix Addition and Subtraction:** Both matrices must have the same dimensions (same number of rows and columns).
  - A = 3 x 3, B = 3 x 3, C (A + B) = 3 x 3 (same dimension)
- **Matrix multiplication:** The number of columns of the first matrix must equal the number of rows of the second matrix.
  - A = 3 x 3, B = 3 x 3, C (A + B) = 3 x 3
- **Matrix Inversion:** can be done for square matrix only having same number rows and columns

Matrix multiplication

Let $A = \begin{bmatrix} 2 & 3 & -4 \\ 1 & 2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 1 \\ -2 & 2 \\ 5 & -3 \end{bmatrix}$. Then

$$AB = \begin{bmatrix} (2)(3) + (3)(-2) + (-4)(5) & (2)(1) + (3)(2) + (-4)(-3) \\ (1)(3) + (2)(-2) + (3)(5) & (1)(1) + (2)(2) + (3)(-3) \end{bmatrix}$$

$$= \begin{bmatrix} -20 & 20 \\ 14 & -4 \end{bmatrix}.$$

Matrices: Exercises

1. Let $\mathbf{A} = \begin{bmatrix} 3 & -2 & 5 \\ 4 & 1 & 2 \end{bmatrix}$, $\mathbf{B} = \begin{bmatrix} 3 \\ -2 \\ 4 \end{bmatrix}$, and

$\mathbf{C} = \begin{bmatrix} 2 & 3 & 4 \\ 5 & 6 & -1 \\ 2 & 0 & 8 \end{bmatrix}$.

(a) What is a_{12}, a_{22}, a_{23}?

(b) What is b_{11}, b_{31}?

(c) What is c_{13}, c_{23}, c_{33}?

(d) List the elements on the main diagonal of C.

Matrices: Exercises

2. Which of the following are diagonal matrices?

(a) $\mathbf{A} = \begin{bmatrix} 2 & 3 \\ 0 & 0 \end{bmatrix}$ **(b)** $\mathbf{B} = \begin{bmatrix} 3 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 5 \end{bmatrix}$

(c) $\mathbf{C} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

(d) $\mathbf{D} = \begin{bmatrix} 2 & 6 & -2 \\ 0 & -1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

(e) $\mathbf{E} = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$

Matrices: Exercises

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 3 \\ 4 & 1 & -2 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 1 \\ 1 & 2 \\ 2 & 3 \end{bmatrix},$$

$$\mathbf{C} = \begin{bmatrix} 1 & -2 & 3 \\ 4 & 2 & 5 \\ 3 & 1 & 2 \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} -3 & 2 \\ 4 & 1 \end{bmatrix},$$

$$\mathbf{E} = \begin{bmatrix} 3 & 2 & -1 \\ 5 & 4 & -3 \\ 0 & 1 & 2 \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} -2 & 3 \\ 4 & 5 \end{bmatrix}.$$

8. If possible, compute each of the following.

(a) $\mathbf{A}^T$ and $(\mathbf{A}^T)^T$

(b) $(\mathbf{C} + \mathbf{E})^T$ and $\mathbf{C}^T + \mathbf{E}^T$

(c) $(\mathbf{AB})^T$ and $\mathbf{B}^T\mathbf{A}^T$ **(d)** $(\mathbf{B}^T\mathbf{C}) + \mathbf{A}$

Matrices: Exercises

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0.2 & 0.4 & 0.2 \\ -0.4 & 0.2 & 0.6 \\ 0.2 & -0.6 & 0.2 \end{bmatrix},$$

$$\mathbf{C} = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad \textit{and} \quad \mathbf{D} = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 0 & -1 \\ 1 & 1 & 1 \end{bmatrix}.$$

27. **(a)** Verify that $\mathbf{C}$ is an inverse of $\mathbf{A}$.

(b) Verify that $\mathbf{D}$ is an inverse of $\mathbf{B}$.